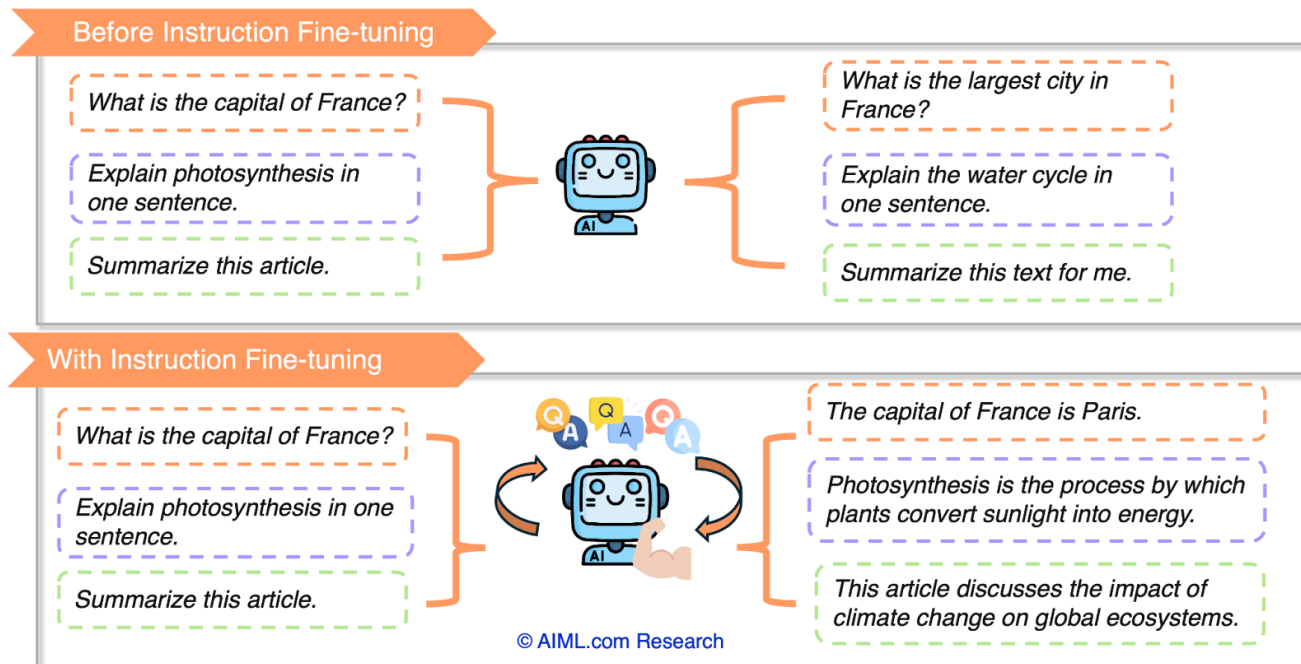


Instruction Fine-Tuning (SFT): Learning to Follow Command

- Pre-trained models are essentially "document completers" that predict the next token based on a massive corpus.
- **Supervised Fine-Tuning (SFT)** teaches the model to act as an assistant by training it on specific {Instruction, Response} pairs.
- This process enables the model to understand the context of a user's command and provide a direct, helpful answer.

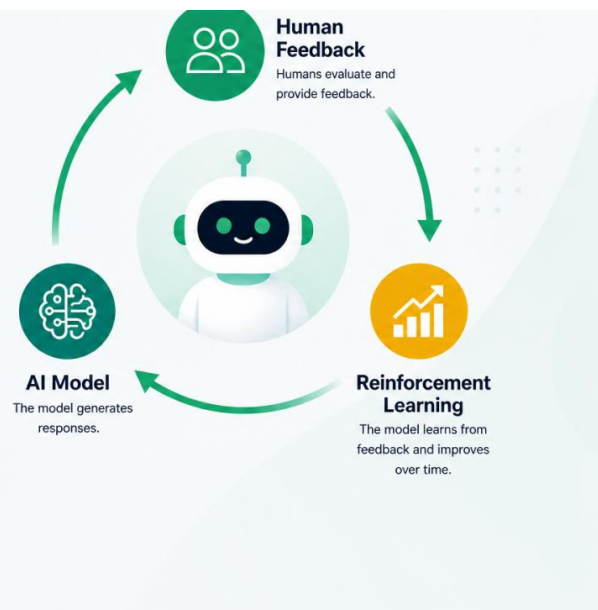
Effectiveness of Instruction Fine-Tuning



RLHF

Reinforcement Learning from Human Feedback

- ✓ Better alignment.
- ✓ Better responses.
- ✓ Better experiences.



RLHF: Making AI Safe and Helpful

RLHF stands for Reinforcement Learning from Human Feedback.

Humans rank multiple model-generated responses to help the model understand which answers are better or safer.

This feedback trains a "Reward Model," which is then used to optimize the LLM's behavior so it aligns with human values.

LoRA: Efficient Tuning on Local Hardware

PEFT (Parameter-Efficient Fine-Tuning) allows for tuning massive models without updating all billions of parameters.

LoRA (Low-Rank Adaptation) works by freezing the original pre-trained weights and adding small, trainable "adapter" matrices.

This drastically reduces memory requirements, making it possible to fine-tune a giant model on a cheap, local GPU.

768

768

